CNENGR202 Instrumentation and Control

Part 2 – L5 Light SONAR Velocity

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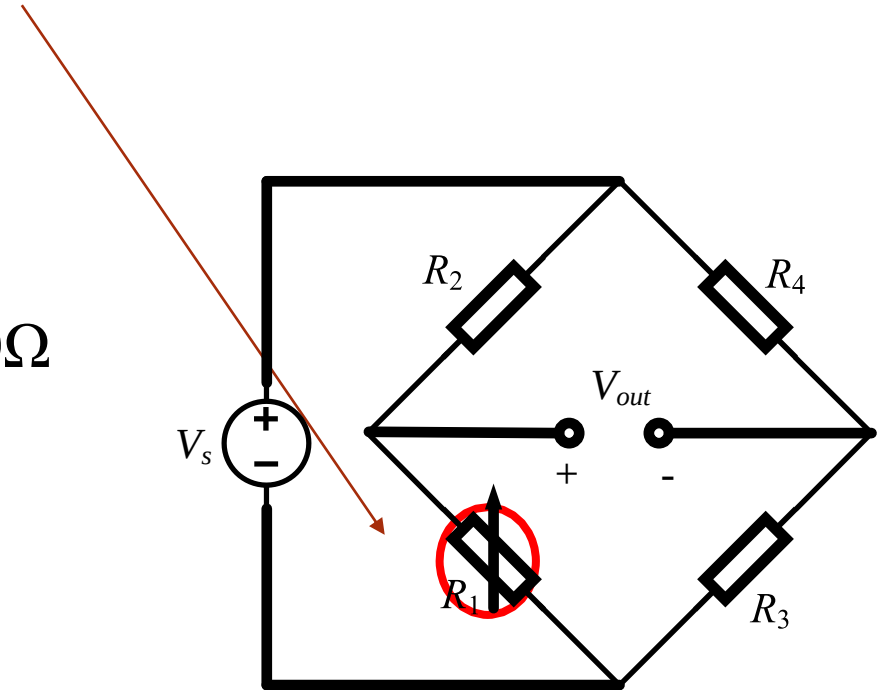
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Home question 1

- Attaching one end of your heavy rope to the wall and a strain gauge half way along the rope (and assume it sticks). This strain gauge goes into a wheatstone bridge circuit and then onto a voltmeter for reading. If the following information is available, how much force are you all producing between you?

- Nominal resistance of strain gauge: 100Ω
- Resistance of other resisters in Wheatstone bridge: 100Ω
- Voltage into Wheatstone bridge: 1V
- Maximum Voltage out of Wheatstone bridge: 1mV
- The value of 1N force cause change 0.1Ω in resistor

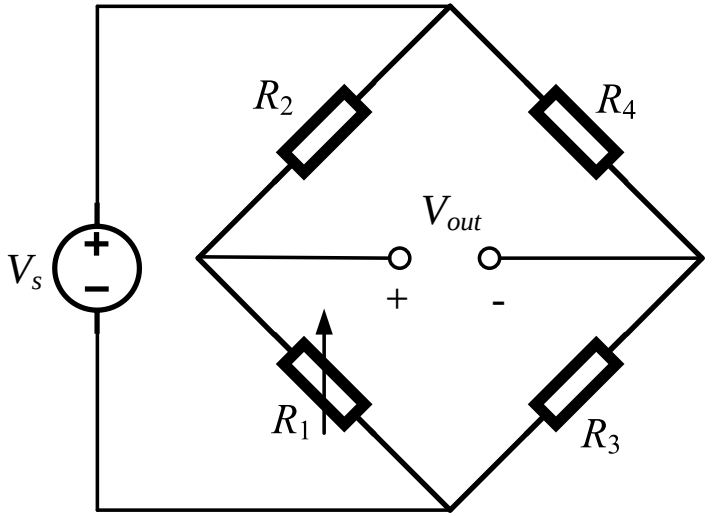


Answer

$$V_{out}/V_{in} = R_1/(R_1+R_2) - R_3/(R_3+R_4)$$

$$1/1000 = (100+\Delta R)/(100+\Delta R+100) - 100/(100+100)$$

$$\Delta R = 0.4\Omega \rightarrow F = 4N$$



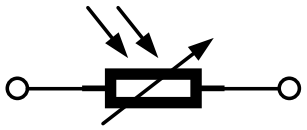
Photoresistive sensor-1

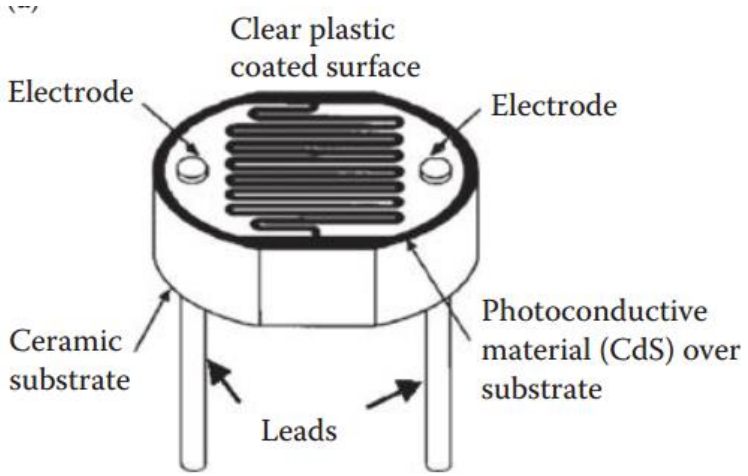
- (Light Dependent Resistor, photoconductor or photocell) – resistance decreases with increasing incident light intensity.

- Made of a high-resistance semiconductor, if light incident on the device is of high enough frequency. Photons absorbed by the semiconductor give bound electrons enough energy to jump into the conduction band. The resulting free electron conduct electricity, thereby lowering resistance.

Characteristics of PGM1200 CdS Photoresistor

Ambient temperature	−30°C ~ +70°C
Spectral peak	560 nm
Photo (light) resistance (at 10 lux)	2 kΩ ~ 5 kΩ
Dark resistance	1.0 MΩ
Response time (rise/decay)	30 ms/40 ms
Dimension (diameter)	5 mm





Photoresistive sensor-2

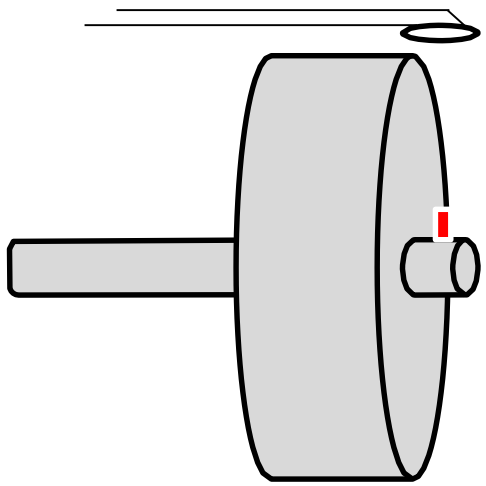
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ISL2902 CdS Photoresistor Datasheet

Maximum power dissipation at 30°C	250 mW
Maximum current	75 mA
Maximum peak voltage	320 V
Typical electrical resistance with illumination by tungsten	400 Ω (at 1000 lux)
lamp of 2854 K at an ambient temperature of 25°C	9000 Ω (at 10 lux)
Dark resistance	>1 MΩ
Rise time: From darkness to 1000 lux	2.8 ms
From darkness to 10 lux	18 ms
Fall time: From 1000 lux to 10-fold resistance	48 ms
From 10 lux to 10-fold resistance	120 ms

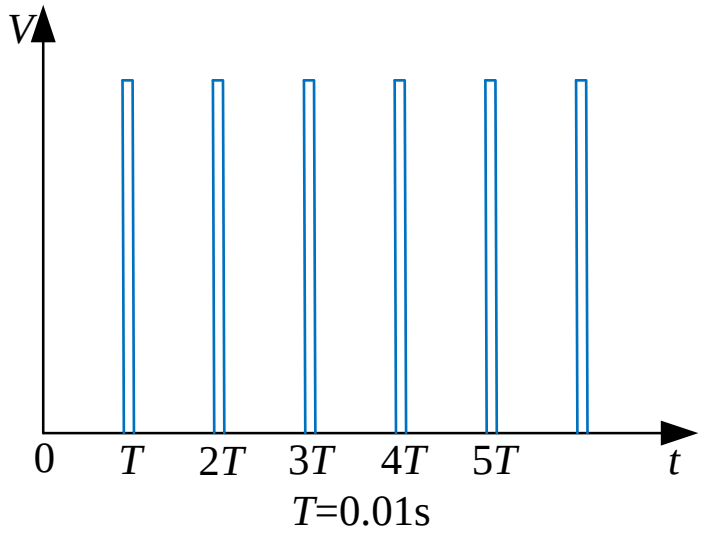
Example 1: Simple application – Speed measurement



Light dependent resistor fixed to chassis

Red LED fixed to axle

- If each wheel has a radius of 10cm, the output looks like the graph below:
  - What is the speed of the car assuming it to be constant?



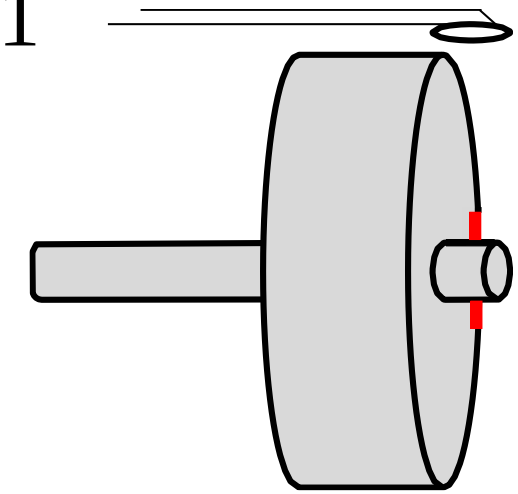
Answers

Radius $$R = 10 \text{ cm} = 0.1 \text{ m}$$

Perimeter $$s = 2\pi R = 2 \times 3.14 \times 0.1 = 0.628 \text{ m}$$

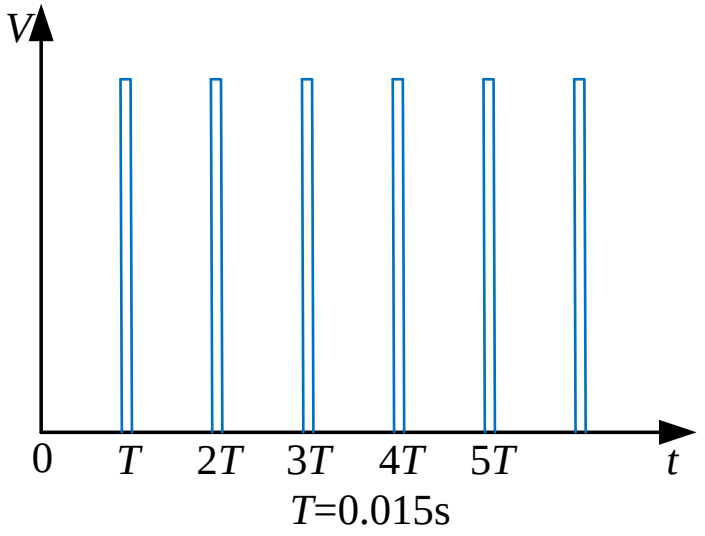
Speed $$v = \frac{s}{T} == \frac{0.628}{0.01} = 62.8 \text{ m/s}$$

Class Question 1



Two red LED fixed to axle

- If the wheel has a radius of 15cm, the output looks like the graph below:
  - What is the speed of the car assuming it to be constant?



Class Question 2

- Design a circuit to realise above function counting the time

Solution

- To turn on the TIMER, the transistor must be ON and OFF, which requires a GATE voltage V_G greater or equal to 5 V:

$$V_G = V_G - 0 \geq 5\text{V}$$

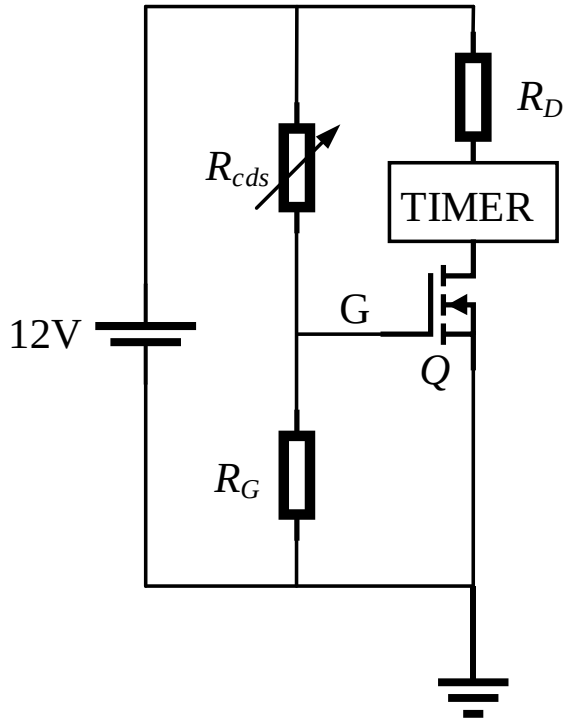
- Assume $R_{cds} = 1\text{k}\Omega$ For lighting

  $R_{cds} = 500\text{k}\Omega$ For dark

$$\frac{R_G}{R_{cdsL} + R_G} \times 12 > 5 \rightarrow R_G > 714\Omega$$

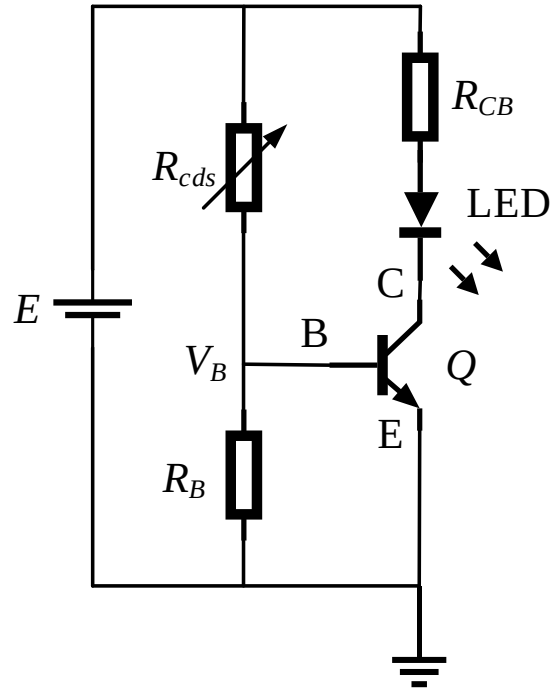
$$\frac{R_G}{R_{cdsD} + R_G} \times 12 < 5 \rightarrow R_G < 357\text{k}\Omega$$

  - So, we can select R_G=1k,10k,100k,200kΩ, ……



Class Question 3

The right figure is a light-controlled LED circuit. It will turn the LED ON when the photoresistor is exposed to light, or turn the LED OFF when the photoresistor is blocked from the light. The control mechanism is the variation of the transistor's base voltage V_{BE} or base current I_B. Once the transistor is ON or ACTIVE, the collector current I_C controls the LED. The LED could be replaced with a relay or motor to actuate other circuits or devices.



If the supplied voltage E = 10 V, the base resistor R_B = 7.5 kΩ. Find the on resistor of photoresistor R_{cds} to turn the transistor ON. Assume that the turn-on voltage of the transistor is 0.7 V.

Proximity sensors - 1

- Types
  - Capacitive
    Magnetic
    Inductive
    Photocell
  - Laser rangefinders
  - Sonar
  - Radar
  - Doppler effect
  - Thermal
  - Infrared
  - Ionising radiation

- **Magnetic**

✓ Emits a beam of electromagnetic radiation, looking for changes in the return signal.

✓ High reliability and long life due to absence of mechanical parts and lack of contact between sensor and object.



Proximity sensors - 2

- Different proximity sensor targets demand different sensors



- **Capacitic**

  ✓ A capacitive or photoelectric sensor might be suitable for a plastic target; inductive proximity sensor requires a metal target.

  ✓ Capacitive proximity sensors is that they are unaffected by dust or opaque containers, allowing them to replace optical devices.

  ✓ 10-mm sensing range, 30 mm in diameter, and incorporates a potentiometer to allow fine tuning of the sensing range

Proximity sensors - 3

- **Magnetic**

✓ Emits a beam of electromagnetic radiation, looking for changes in the return signal.

✓ High reliability and long life due to absence of mechanical parts and lack of contact between sensor and object.



E.G. Parking sensors

- Technology that allows the driver of an automobile, truck, or commercial vehicle to be alerted to nearby objects in their path.
- The basic design puts sensors in the vehicle's bumper which send ultrasonic waves at 40,000 Hz that:





- detect obstacles be hind the vehicle
- A controller installed inside the vehicle then receives the signal from the sensors and transmits it to a speaker that emits a tone, or to a combination speaker and display unit.

SONAR (Sound Navigation and Ranging) - 1

- Sound waves are radiated down into the deep ocean, bend back up to the surface in great arcs due to the increasing pressure (and hence sound speed) with depth.
- Active sonar creates a ping, and then listens for an echo.
- This pulse of sound is generally created electronically (signal generator, power amp, electro-acoustic transducer).
- To measure the bearing, several hydrophones are used measuring the relative arrival time at each.

SONAR (Sound Navigation and Ranging) - 2

- Hydrophone: piezoelectric microphone designed to be used underwater for recording or listening to underwater sound.

- A hydrophone can 'listen' to sound in air, but will be less sensitive due to its acoustic impedance matching to water.

  - Frequency: 20Hz - 20kHz
  - Sensitivity: 35dB ± 4dB
  - Voltage - Rated: 2V
  - Current: 500 µA
  - S/N Ratio: 62dB

Character of a sample

